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Identifying mortality risk factors amongst acute coronary syndrome patients admitted to Arabian Gulf hospitals using machine-learning methods

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Abstract

Acute coronary syndrome (ACS) is a leading cause of mortality and morbidity in the Arabian Gulf. In this study, the in-hospital mortality amongst patients admitted with ACS to Arabian Gulf hospitals is predicted using a comprehensive modelling framework that combines powerful machine-learning methods such as support-vector machine (SVM), Naïve Bayes (NB), artificial neural networks (NN), and decision trees (DT). The performance of the machine-learning methods is compared with that of the performance of a commonly used statistical method, namely, logistic regression (LR). The study follows the current practise of computing mortality risk using risk scores such as the Global Registry of Acute Coronary Events (GRACE) score, which has not been validated for Arabian Gulf patients. Cardiac registry data of 7,000 patients from 65 hospitals located in Arabian Gulf countries are used for the study. This study is unique as it uses a contemporary data analytics framework. A *k*-fold (k = 10) cross-validation is utilized to generate training and validation samples from the GRACE dataset. The machine-learning-based predictive models often incur prejudgments for imbalanced training data patterns. To mitigate the data imbalance due to scarce observations for in-hospital mortalities, we have utilized specialized methods such as random undersampling (RUS) and synthetic minority over sampling technique (SMOTE). A detailed simulation experimentation is carried out to build models with each of the five predictive methods (LR, NN, NB, SVM, and DT) for the each of the three datasets *k*-fold subsamples generated. The predictive models are developed under three schemes of the *k*-fold samples that include no data imbalance, RUS, and SMOTE. We have implemented an information fusion method rooted in computing weighted impact scores obtain for an individual medical history attributes from each of the predictive models simulated for a collective recommendation based on an impact score specific to a predictor. Finally, we grouped the predictors using fuzzy c-mean clustering method into three categories, high-, medium-, and low-risk factors for in-hospital mortality due to ACS. Our study revealed that patients with medical history related to the presences of peripheral artery disease, congestive heart failure, cardiovascular transient ischemic attack valvular disease, and coronary artery bypass grafting amongst others have the most risk for in-hospital mortality.

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KEYWORDS

acute coronary syndrome (ACS), decision tree, fuzzy c-mean clustering, Global Registry of Acute Coronary Events (GRACE) risk score, imbalance data, information fusion, logistic regression, machine learning, mortality, Naïve Bayes, neural networks, predictive analytics, support-vector machine

1 | INTRODUCTION

Acute coronary syndromes (ACS) are a widely encountered group of cardiovascular diseases (Elsaesser & Hamm, 2004; Théroux & Fuster, 1998). ACS include unstable angina, non-ST-segment elevation myocardial infraction (NSTEMI), and ST-segment elevation myocardial infraction (STEMI). ACS are linked with significant negative risk events including mortality, morbidity, recurrent ischemia, and bleeding complications. Nevertheless, studies reveal that these short- and long-term mortality risks associated with ACS vary with the patient's condition and medical history. This underlines the need to conduct improved early risk stratification to identify the patients at a higher risk of mortality during the time of admission. Such stratification helps appropriate treatment to be designed for mitigating the risk of adverse outcomes like in-hospital mortality at the time of admission.

Traditional methods based on single marker of risk such as blood levels of cardiac troponin levels in blood may not be accurate enough to predict outcome for clinical success (Steg, FitzGerald, & Fox, 2009). Therefore, numerous risk stratification methods have been developed that consider multiple determinants derived from randomized clinical trials or registry data. Further enhancement of accuracy through more detailed clinical assessments and sophisticated machine learning (ML) can enable superior prediction of the likelihood of negative mortality-related risk events (Ramsay, Podogrodzka, McClure, & Fox, 2007). Risk models derived from randomized trials like the study by Antman et al. (2000) on thrombolysis in myocardial infarction offer the advantage of greater robustness of data. However, these risk models are based on a single selected patient cohort, and the generalization of the results to the wider patient population may lead to erroneous results. On the contrary, models derived from studies conducted on registry such as Global Registry for Acute Coronary Events (GRACE) are more reliable because the samples are large and randomly selected. These data widely collected across many different countries can often be associated with higher external validity.

The two primary GRACE risk scores currently available predict the risk of in-hospital mortality and "6-month post-discharge mortality." These two scores are a sum of individual scores pertaining to several clinical factors relevant to the outcome (mortality) including age, gender, heart rate (HR), blood pressure, congestive heart failure, renal impairment, ST segment elevation on the electrocardiogram findings, cardiac arrest, and rise in troponin levels. The "6-month post-discharge mortality" GRACE risk score is validated amongst European populations by Abu-Assi et al. (2000). Alnasser et al. (2015) demonstrate that the score can also be useful in predicting mortality up to 2-year post discharge. However, this score needs validation amongst patients in the Arabian Gulf because risk factor profiles and medical management may vary significantly from region to region. Therefore, the current study is designed to validate the GRACE risk score in predicting mortality at 1-year post discharge amongst Arabian Gulf patients. Thalib et al. (2017) noted that GRACE has not been studied in connection with Arabian Gulf patients for propensity (survival) analysis and conduct one of the first studies with Arabian Gulf GRACE dataset. Granger et al. (2003) analyse in-hospital mortality for GRACE data for hospitals in Western Europe, Oceania, and North America. However, many of these studies employ traditional approaches such as logistic regression (LR) for in-hospital ACS mortality prediction. Despite the popularity of the method, the LR method and other traditional methods have shortcomings owing to the assumptions associated with these methods.

This paper seeks to remedy this problem by proposing a unified and unbiased decision support tool that utilizes multiple novel predictive models from ML through an efficient information collection and fusion process. The motivation underlying this study is to exploit the advantages of advanced tools from ML and predictive analytics to develop a comprehensive decision support tool that classifies predictors into risk categories, that is, high-, medium-, and low-risk predictors. The remainder of the paper is organized as follows: a brief literature review is followed by a detailed description of the methodology in Section 2. In this section, three main components are discussed. First, details of data processing and cleaning are presented. Second, data is sampled using 10-fold cross-validation is discussed. Third, the imbalanced data management is performed owing to the lower number of in-hospital mortality records compared with the survival records. Last, five predictive models (LR, support-vector machine [SVM], Naïve Bayes [NB], artificial neural networks [NNs], and decision trees [DT]) are developed. In Section 4, the predictive models developed earlier in Section 3 are first simulated and later assessed based on well-established performance measures (area under curve [AUC], accuracy, sensitivity, and specificity). We have utilized training datasets for building (training) models and later assessed model using the corresponding validation datasets. Brief descriptive analytics are also presented in Section 4.1. In Section 5, a comprehensive methodology is implemented to compute collective impact of each of 24 predictors, which is referred to as process of information fusion (IF; Dag, Oztekin, Yucel, Bulur, & Megahed, 2017). Later in Section 5, the predictors are classified into three risk groups (high, medium, and low) using a fuzzy c-mean clustering approach. The findings and the impact of this study in contrast to the published research is also discussed in Section 5. Finally, the conclusions are drawn and the suggestions for the future work are articulated in Section 6.

2 | METHODOLOGY

The comprehensive data analytic methodology proposed in this paper is outlined in Figure 1. The methodology consists of six sequential stages. The first three stages (Stages I, II, and III) are related to data preparation, data subsampling for cross-validation, and data imbalance management. In Stage I, we preprocessed the dataset consisting of 284 variables to screen 24 potential predictors associated with the medical history (see Appendix A) and the general demographics (age, gender, country, and etc.) of patients. A *k*-fold cross-validation procedure using 10 folds commonly adopted by many of the studies is employed in Stage II (Dag et al., 2017). Each of *k*-fold, where $k = 10$, samples are divided into training dataset and the validation (holdout) datasets using a random selection in the proportion 80:20. The purpose of this task is to avoid overfitting, critically assess the performance of the prediction models, and rank the models along with the predictors based on their performance measures. Further, due to the low occurrence of in-hospital moralities, 319 out of the total 6,847 patients in the dataset, the data are imbalanced. This problem is resolved using data imbalance management techniques shown in Stage III in Figure 1. Stage IV, utilizes the data prepared in earlier stages to build the predictive models. Stage V is made up of two substages: V(A) and V(B). In V(A), the predictive models are assessed using the standard prediction performance in terms of the fitness of the models on the train dataset and later the validation dataset. In the substage V(B), the IF technique integrates all predictive models to collectively identify the influential predictors. Finally, Stage VI involves the development of a fuzzy-clustering technique for grouping these predictors screened by Stage VB, into clusters ranging from "the most influential" to "the least influential". After the application of the six-stage model, recommendations resulting from the analysis for healthcare practitioners and professionals are listed.

2.1 | Data processing and cleaning

In this section, we detail the processing and cleaning done in Stage I of our proposed methodology. The data were collected from a second Gulf Registry of Acute Coronary Events (Tang, Wong, & Herbison, 2007; Thalib et al., 2017) including consecutive patients with ACS from six Arabian Gulf Countries (Bahrain, Saudi Arabia, Qatar, Oman, United Arab Emirates, and Yemen). These patients belonged to 65 different hospitals during the period of data collection during the period October 2008 to June 2009. The patients in the dataset were diagnosed with one of the following conditions: unstable angina, non-ST-segment elevation acute coronary syndrome (NSTEACS), NSTEMI (included as NSTEACS), and STEMI. Diagnosis of NSTEACS and definitions of data variables were based on the American College of Cardiology clinical data standards (see Antman et al., 2000, for definitions of NSTEACS, NSTEMI, STEMI, and ACS). The study received ethical approval from the institutional ethical bodies in all participating countries. The dataset contains information related to the patient demographics including, age, gender, marital status, medical history, amongst others. The data were deidentified to preserve the anonymity of patients for the purposes of this study. The dataset was very rich and consisted of over 284 information fields. Authors consulted medical experts for determining the most essential data attributes in order

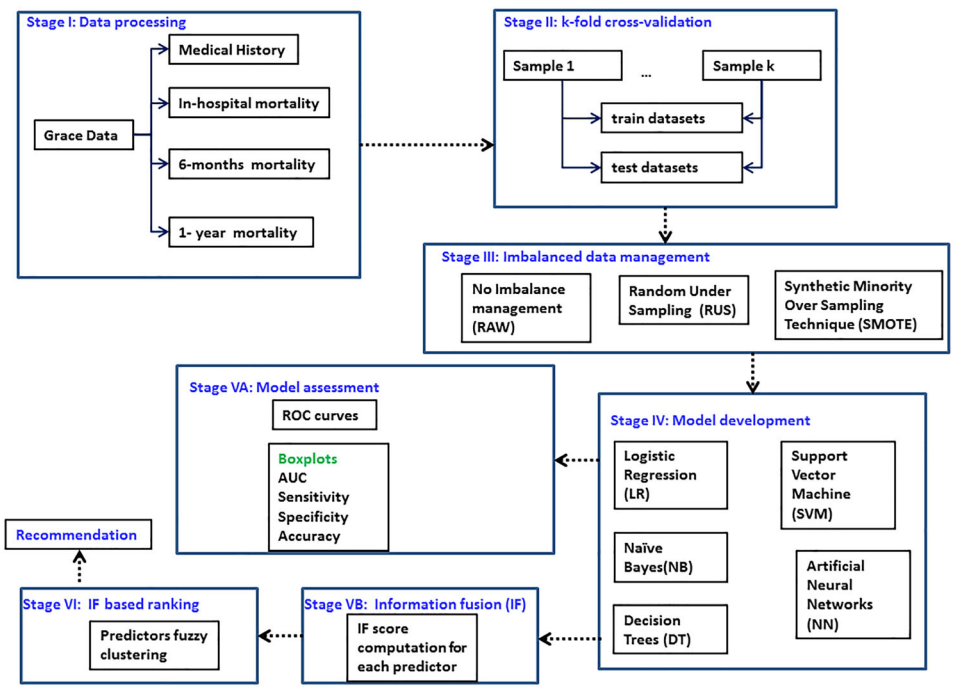

FIGURE 1 Proposed data process and analytics scheme

to reduce the fields. Only 24 medical history related fields were finally retained in addition to the demographic data that contained essential information such as (a) cardio medical history (angina, MI, congestive heart failure [CHF], percutaneous coronary intervention [PCI], coronary artery bypass grafting [CABG], stroke, and peripheral artery disease [PAD]), (b) cardiovascular risk factors (smoking history, diabetes, hypertension, and hyperlipidemia), (c) presentation characteristics (Heart Rate (HR), Systolic blood pressure (SBP), Diastolic Blood Pressure (DBP), and Killip class type), and (d) auxiliary tests results (elevated troponin [CK], ECG-ST elevation see Table 2). A detailed description of these medical history related predictor variables are included in the Appendix A. The original dataset had over 7,500 medical records; however, there were cases in which the shortlisted 24 field information was missing. Such cases were less than 10%, these records with missing information were removed from the next stage of the analysis. We followed a standard procedure suggested in R-package that is to plot the heatmap for these 24 attributes and then remove the missing value records. This resulted the processed dataset containing 6,847 (n = 6,847) observations out of which 319 (n = 319) cases of in-hospital mortality (i.e., Alive = 0 or Death = 1) were observed.

2.2 | Data sampling

The preprocessed dataset obtained subsequent to the application of the procedure described in Section 4.1 is sampled using 10-fold cross-validation and later treated for data imbalance management. Figure 2 shows a schematic framework.

2.2.1 | *K*-fold cross-validation subsampling

As shown in the proposed framework in Figure 1 (see Stage II), a robust data subsampling scheme is developed. The 10-fold cross-validation procedure widely used in many related studies (Dag et al., 2017; Dag, Topuz, Oztekin, Bulur, & Megahed, 2016) is utilized. The proposed subsampling technique uses k-fold (k = 10) cross-validation approach to randomly but comprehensively sample the total observations (n = 6,847) into 10 diverse samples. Each sample is randomly divided into train (n ≈ 4,240) and validation (n ≈ 472) datasets. This task is completed using an existing R package for cross validation called "cvTools" (Team, 2013). A detailed schematic framework for 10-fold cross-validation is presented in Figure 2.

2.2.2 | Imbalance data treatment

The GRACE dataset used for the current study, contains a total of 6,847 patient records in which the in-hospital mortalities are a very low number, only 319. Thus, the GRACE dataset requires application of methods to manage imbalance (Dag et al., 2017; Kotsiantis, Kanellopoulos, & Pintelas, 2006). This data imbalance problem is addressed via the application of the random undersampling (RUS) and the synthetic minority oversampling technique (SMOTE; Chawla, Bowyer, Hall, & Kegelmeyer, 2002) on each of the k-folds of the training datasets. This task was completed in Stage III of data analytic framework in Figure 1. The details on this is presented Figure 2. Ten (10) training samples are assigned to each of the imbalance methods, RUS and SMOTE, and ten training samples are run without any imbalance method. Thus, in total, 30 training samples are used to build ML models (LR, SVM, NN, NB, and DT; see Figure 2).

3 | PREDICTIVE MODELS DEVELOPMENT

In this paper, we have implemented four well-known ML-based data analytics models, artificial NN, NB, SVM, and DT, and the conventional statistical method, LR. The selection of these ML methods is based on the evidence of superior performance with respect to classification problems

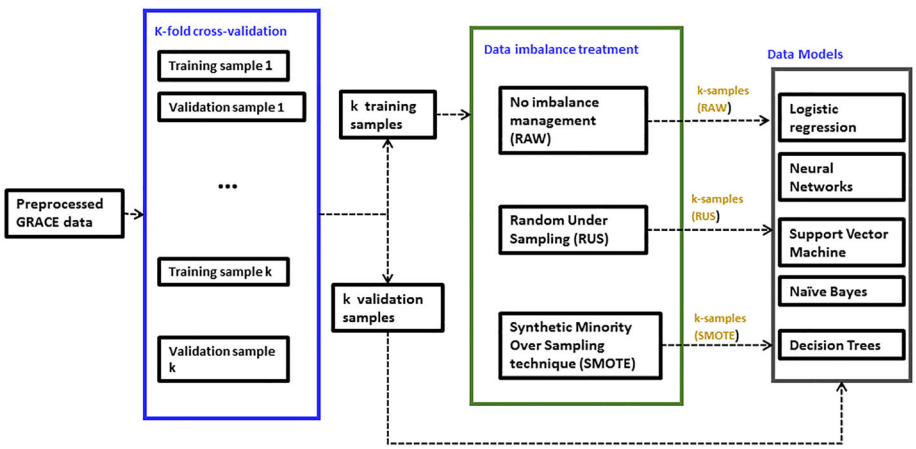

FIGURE 2 Data k-fold sampling and imbalance management scheme

often encountered in health-care research (Del Rizzo et al., 1999; Shipp et al., 2002; Oztekin et al., 2009; Oztekin, Kong, & Delen, 2011; Deo, 2015; Dag et al., 2016; Myers, Scirica, & Stultz, 2017). The preliminary simulation results of these ML methods have been encouraging and motivated the use of these methods for this study. Many examples of the application of ML are existing in medical literature (Kononenko, 2001; Cleophas, Zwinderman, & Cleophas-Allers, 2013; Deo, 2015; Obermeyer & Emanuel, 2016). A detailed coverage of the ML methods tested is out of the scope of this paper; however, details on these methods can be referenced from studies such as, Kononenko (2001), Waljee and Higgins (2010), and Alpaydin (2014) amongst others. In the following Subsections 3.1 to 2.3.5, the ML methods and LR are briefly described.

3.1 | Logistic regression

LR is a widely utilized technique to classify outcomes into likely categories (Hand & Henley, 1997). It is regarded as the standard for predicting binary, binomial, and multinomial categorical outcomes. The predicted outcomes are discrete, and therefore, the generally existing linear regression modelling approaches may not directly achieve a model fit to the data. The LR predicts the odds that an outcome is dependent on single or multiple predictors. More commonly encountered two-class problem are often classified with an odd greater than 50% into the desired event (commonly coded as "1"). Any odd less than or equal to 50% is contemplated into non-event (that is commonly coded as "0"). Even though LR is a very powerful modelling tool, it assumes that the log of odd (commonly regarded as logit) is linearly related to its predictors. LR can be viewed as a special case of linear regression. The main distinguishing component is that the outcome variable is categorical (often dichotomous) in many applications, where in linear regression, this outcome can be any real number (Hosmer & Lemeshow, 1989).

3.2 | Artificial NNs

An artificial neuron constitutes the basic unit in an NN. The artificial neuron is a simplistic representation that mimics the signal integration and threshold firing behaviour of biological neurons using the mathematical formulations (Sordo, 2002). The information flow between artificial neuron is computed via connections between peer neurons. The NNs are widely used for various classification and prediction problems due to their ability to handle both linear and nonlinear functions for modelling data. This feature greatly empower modellers to obtain superior analyses when the historical data is non-linear. The applications of NN can be found in numerous domains of physical and social sciences. Bishop (1995) reports a large number of real-life problems solved using NN. Similar to several other ML methods, NN method is also referred to as a "black box" classifiers; that is, the rationale for the outcome is not given. In this paper, we have used a multilayer NN the details of which can be learnt by referring to Han, Pei, and Kamber (2011) and Neural Network Toolbox, MATLAB (2015).

3.3 | Support-vector machine

SVM was first proposed by Cortes and Vapnik (1995) for classification problems. It is extremely powerful because they can be used for linearly and nonlinearly separable datasets (Han et al., 2011). SVM determines the optimal hyper-plane amongst the classes by optimizing the marginal differences between them. The data points that are lying on the boundaries are described as support vectors. The middle of the margin is called the optimal separating hyper-plane. Like other classification techniques in ML, SVM is also subject to the criticism of being a "black box" technique. SVM is also computationally more expensive compared with the other statistical comparable methods.

In contrast to artificial NN and DT training process, SVM training process minimizes the associated risks (Vapnik, 1998). Minimization of the empirical risk stands for reducing the classification error on a training dataset. Whereas, minimization of structural risk is to mitigate the classification error on unobserved patterns. Amongst the existing ML algorithms, most only minimize the empirical risk; therefore, SVM works on two aspect on minimizing both the empirical and structural risks simultaneously unlike most ML algorithms.

3.4 | Naïve Bayes

NB is a widespread ML tool for classification. NB is very simple with a very high computational efficiency, whilst it maintains a good classification accuracy. It very well suited for high dimensional data such as texts and so forth. Amongst the shortcomings for NB is the underlying assumption of strong independence between attributes. This can substantially deteriorate the classification performance of NB. To address this shortcoming, various improvements are suggested to NB. The improvements embed approaches such as structure extension, attribute selection, attribute weighting, instance weighting, local learning, amongst others.

3.5 | Decision tree

DT are amongst the most understandable and very intuitive to interpret data analytics methods. That is why DT are used in many data mining and predictive applications for medical sciences (see Safavian and Landgrebe (1991); Kusiak, Dixon, & Shah, 2005; Oztekin et al., 2009; Nakayama et al., 2012). DT model is primarily a directed acyclic graphical model. Although DT is proposed for the binary classification, as directed (rooted) tree, the DT is required to meet three essential properties: (a) There must only one in the DT that has no edges entering into it, and this node is regarded as the root; (b) every node with the exception of the root has exactly one edge entering it; and (c) there exists a unique path from the root to each node. A more detailed discussion on DT is presented in Safavian and Landgrebe (1991).

4 | SIMULATION RESULTS OF ML MODELS

On the basis of methodology framework presented in Section 2, the five predictive models (LR, NN, NB, SVM, and DT) were implemented in MATLAB to predict the effect of the patients medical and other related conditions on the propensity for in-hospital mortality (death). Prior to this implementation, the descriptive analysis of preprocessed data is presented in Section 4.1.

4.1 | Descriptive analytics

Referring to Table 1, a total of 6,847 patients in the dataset consisted of 319 in-hospital mortalities. That is, the mortality rate is 4.66% (<5%). The highest percentage of in-hospital mortality is observed in Yemen (46%), followed by Oman (28%).

Table 2 outlines the basic characteristics of the GRACE cohort outlining the fundamental demographic characteristics, cardiovascular risk factors, medical history, and some auxiliary test outcomes.

4.2 | Performance analysis and comparison

We report the performance of five ML methods. A total of 150 (5 × 3 × 10) models (i.e., five methods [LR, NN, NB, SVM, and DT], three data imbalance management [no management, RUS, and SMOTE], and 10 training samples [10-folds]) are developed.

Figure 3 presents receiver operating characteristic (ROC) curve for a randomly selected 1 out 10 training datasets generated using k-fold, k = 10, cross-validation with no data imbalance management, that is, raw (referred to as RAW) data. The plots also show AUC. It may be noted that ROC plots are obtained for all simulations with RAW, RUS, and SMOTE; however, only one plot out of 30 plots is shown in Figure 3.

Similar to Figure 3, in Figure 4 again ROC curve plots are shown for the simulation run that corresponds to the one that generated Figure 3. The primary difference between the two ROC plots is that the AUC values are lowered for the validation dataset compared with the corresponding values for the training dataset.

In contrast to Figure 3, Figure 4 shows the typical ROC curve plots for all the five prediction models but on the validation dataset for the corresponding training dataset reported earlier Figure 3. The remaining nine ROC curves for each of the training and validation datasets resemble the curves shown in Figures 3 and 4, respectively, and are therefore not presented. Authors have also observed that ROC curve plots considerably vary from one simulation to another. Naturally, AUC for each of the ML methods for the training sample is superior to the corresponding AUC for respective models for the validation sample. This is widely observed for all schemes for sample, that is, RAW, RUS, and SMOTE.

Table 3 reports the summary (mean and standard deviation) of the performance measures calibrated for each of the ML models (LR, NN, NB, SVM, and DT) under the three distinct data sampling strategies. These data sampling strategies include no data imbalance management, referred to as RAW, RUS, and SMOTE. The performance measures included, accuracy, sensitivity, and specificity. These measures are widely utilized to

TABLE 1 In-hospital mortality statistics country wise

Country	In-hospital mortality, *n* (%)
Saudi Arabia	53 (16.61)
Bahrain	7 (2.19)
Yemen	146 (45.77)
Oman	88 (27.59)
United Arab Emirates	11 (3.45)
Qatar	14 (4.39)
Total	319

TABLE 2 Baseline characteristics of Global Registry for Acute Coronary Events (GRACE) population cohort

Demographic characteristics	Total (*n* = 6,847)
Age, median (IQR)	56 (48–65)
Gender (male), *n* (%)	5,409 (79%)
Country, *n* (%)	
Saudi Arabia	1,489 (21.7%)
Bahrain	457 (6.7%)
Yemen	1,745 (25.5%)
Oman	1,909 (27.9%)
United Arab Emirates	580 (8.5%
Qatar	667 (9.7%)
Cardiovascular risk factors, *n* (%)	
Current smoker (Smok)	2,439 (35.62%)
Diabetes (DM)	2,624 (38.32%)
Hypertension (Hyp)	3,207 (46.84%)
Hyperlipidemia (Hyplip)	2,232 (32.60%)
Previous history, n(%)	
Angina	2,586 (37.78%)
Myocardial infarction (MI)	1,277 (18.65%)
Congestive heart failure (CHF)	472 (6.89%)
Percutaneous coronary intervention (PCI)	623 (9.10%)
Coronary artery bypass grafting (CABG)	285 (4.16%)
Transient ischemic attack (TIA)	301 (4.40%)
Peripheral artery disease (PAD)	129 (1.88%)
Presentation characteristics	
Heart rate (HR), median (IQR), beats/min	84 (97–72)
Systolic blood pressure SBP, median (IQR), mm Hg	135 (154–120)
Diastolic blood pressure (DBP), median (IQR), mm Hg	80 (92–70)
Killip, Class I, *n* (%)	5,204 (76%)
Killip, Class II, *n* (%)	1,022 (14.93%)
Killip, Class III, *n* (%)	378 (5.52%)
Killip, Class IV, *n* (%)	243 (3.55%)
Auxiliary tests	
Elevated troponin or CK *n* (%)	4,843 (70.73%)
Electrocardiography (ECG)-ST elevation, *n* (%)	4,368 (63.8%)

assess the performance of predictive models (Dag et al., 2017; Kotsiantis et al., 2006). Boxplots are employed to compare the five ML models with respect to four performance measures AUC, accuracy, sensitivity, and specificity. Figures 5–8 present boxplots for the five predictive models LR, NN, NB, SVM, and DT simulated in this study with the three different management schemes, that is, RAW, along with RUS and SMOTE.

It very essential to observe here that specificity that measures particularly the correct prediction of the mortality amongst the patients is the most desired performance measure to improve in this study. The proposed approach to data imbalance management has considerably improved the specificity for most learning methods (NN, NB, SVM, and DT), which is observable from Figure 8. With some ML models, RUS is found better in improving the specificity measure, whereas SMOTE also helped some ML models enhance specificity measure. Table 3 reports a comparison of these measures.

It may also be noted that the accuracy has not been improved significantly by using the two data imbalance management techniques, RUS and SMOTE. Typically, as the accuracy of predicting the minority class (percentage of mortalities), that is, specificity, increases, the overall accuracy may not improve. Thus, there is a likelihood that by removing majority samples in RUS and adding minority samples in SMOTE overall accuracy

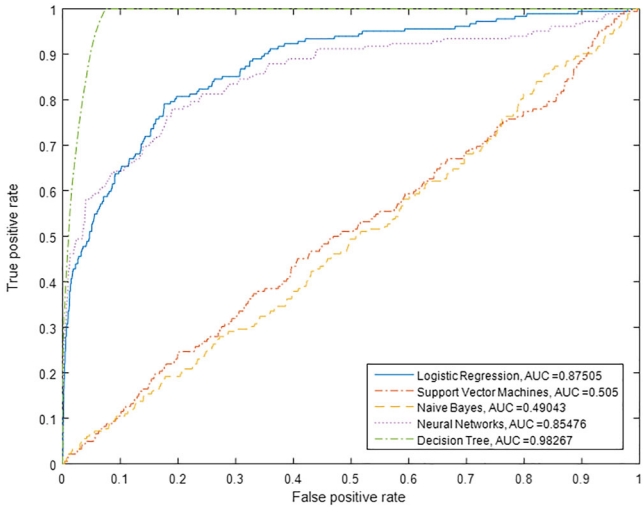

FIGURE 3 An arbitrary receiver operating characteristic curve plot for training data set using RAW sample

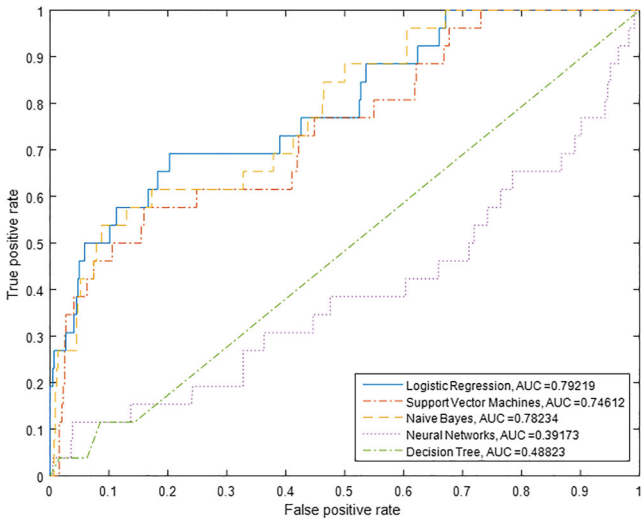

FIGURE 4 An arbitrary receiver operating characteristic curve plot for validation data set using RAW sample

is compromised for the sake of achieving the aim of the study of more accurate prediction of mortality amongst patients. Moreover, predictive accuracy often employed to evaluate performance of a classifier might not be the appropriate measure when the data is imbalanced. Similarly, sensitivity is found to be higher for the RAW, and SMOTE achieves higher sensitivity values compared with RUS for all methods except NB.

5 | DISCUSSION

In this section, we discuss the results in conjunction with the IF method. IF is a technique that systematically combines information gathered from multiple predictive models to mitigate prediction bias (Torra, 2003). The IF primary objectives are to mitigate the model's precariousness and improve the ability to obtain the information with an increased confidence when compared with that provided by a single model. Studies have shown that using IF, the prediction bias can be reduced whilst information accuracy can be enhanced (Cang & Yu, 2014; Graefe, Armstrong, Jones, & Cuzán, 2014). There is a growing interest to use IF approach to ML due to its robustness (Dag et al., 2017).

Next, we perform an IF-based sensitivity analysis. Similar to Dag et al. (2017), only the SMOTE training data samples are used mainly due to their well-balanced structure. In Equation (1), methodology of IF score computation is mathematically formulated for a predictor. Here, IF score is computed for each of the predictors ($i = 1,\cdots,I$), where $I = 24$ (24 predictors) in this study. Here, K ($K = 10$, 10-folds SMOTE samples) represents the number of replications performed. For each predictor, i, and sample replication, k, the sensitivity ($s_{i\ k}$) is computed along with corresponding AUC score, AUC_{ik}. For each predictor, i the cumulative sensitivity (S_i) is determined using Equation (1), which is regarded as IF score for the predictor, i.

TABLE 3 Summary of predictive performance measures

Performance Measure	Method	RAW Mean (SD)	RUS Mean (SD)	SMOTE Mean (SD)
AUC	LR	0.847 (0.035)	0.832 (0.038)	0.843 (0.033)
	NN	0.449 (0.081)	0.760 (0.040)	0.746 (0.069)
	NB	0.830 (0.031)	0.818 (0.037)	0.832 (0.030)
	SVM	0.677 (0.074)	0.831 (0.044)	0.840 (0.037)
	DT	0.517 (0.039)	0.740 (0.058)	0.602 (0.044)
Accuracy	LR	0.960 (0.008)	0.753 (0.028)	0.782 (0.029)
	NN	0.927 (0.011)	0.760 (0.040)	0.863 (0.028)
	NB	0.907 (0.007)	0.810 (0.044)	0.775 (0.030)
	SVM	0.955 (0.010)	0.771 (0.031)	0.789 (0.024)
	DT	0.916 (0.015)	0.672 (0.031)	0.930 (0.010)
Sensitivity	LR	0.995 (0.004)	0.752 (0.029)	0.763 (0.098)
	NN	0.969 (0.013)	0.673 (0.036)	0.884 (0.031)
	NB	0.928 (0.007)	0.817 (0.049)	0.777 (0.035)
	SVM	0.998 (0.002)	0.773 (0.033)	0.791 (0.024)
	DT	0.956 (0.011)	0.670 (0.034)	0.962 (0.013)
Specificity	LR	0.179 (0.101)	0.760 (0.116)	0.781 (0.027)
	NN	0.023 (0.034)	0.737 (0.088)	0.417 (0.149)
	NB	0.447 (0.122)	0.630 (0.167)	0.750 (0.080)
	SVM	0.023 (0.058)	0.741 (0.088)	0.743 (0.125)
	DT	0.050 (0.041)	0.743 (0.100)	0.242 (0.091)

Abbreviations: AUC, area under curve; DT, decision trees; LR, logistic regression; NB, Naïve Bayes; NN, neural network; RUS, random undersampling; SMOTE, synthetic minority over sampling technique; SVM, support-vector machine.

TABLE 4 Risks level determination using fuzzy clustering

Predictors	IF score	Risk level
PAD	0.984	High
CHF	0.942	High
Val_D	0.838	High
CVA_TIA	0.833	High
CVA_TIA.1	0.833	High
CABG	0.833	High
Killip	0.810	High
Gender	0.794	Medium
C_RF	0.787	Medium
DBP	0.785	Medium
SBP	0.784	Medium
Myocardial infarction (MI)	0.774	Medium
HR	0.685	Medium
Age	0.644	Medium
Diabetes mellitus (DM)	0.621	Medium
Country	0.550	Medium
Hyp	0.460	Medium
Smoke	0.420	Medium
ANGINA	0.359	Medium
Hyplip	0.354	Medium
ECG_ST	0.348	Medium
Troponin	0.244	Low
FH_P_CAD	0.106	Low
PCI	0.097	Low

Abbreviations: CABG, coronary artery bypass grafting; CHF, congestive heart failure; HR, heart rate; IF, information fusion; PAD, peripheral artery disease; PCI, percutaneous coronary intervention.

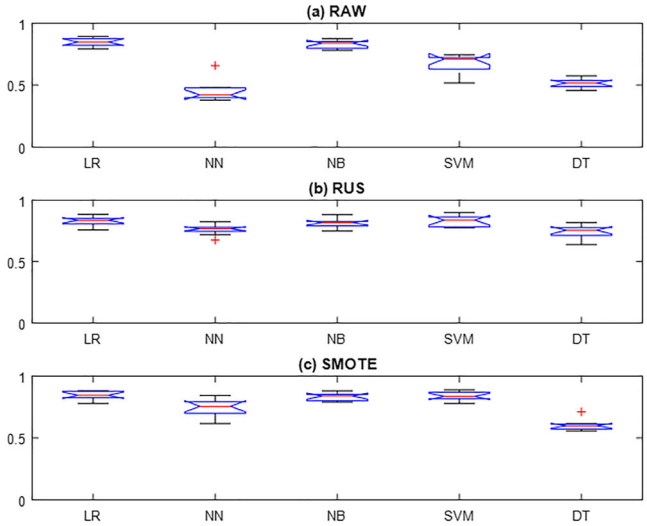

FIGURE 5 Box plots, area under curve comparison

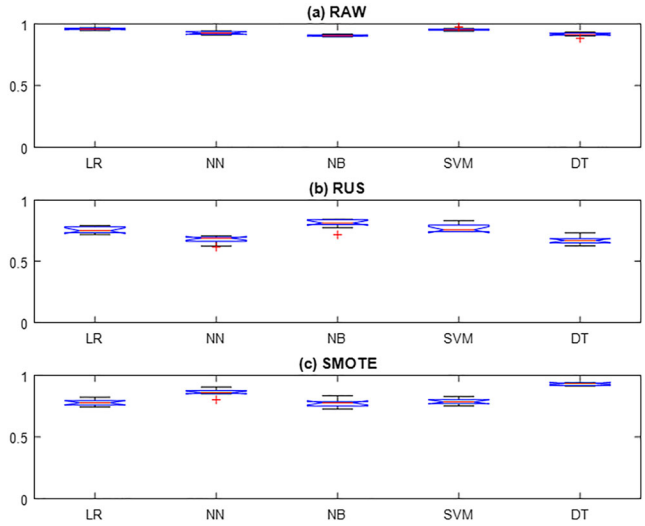

FIGURE 6 Box plots, accuracy comparison

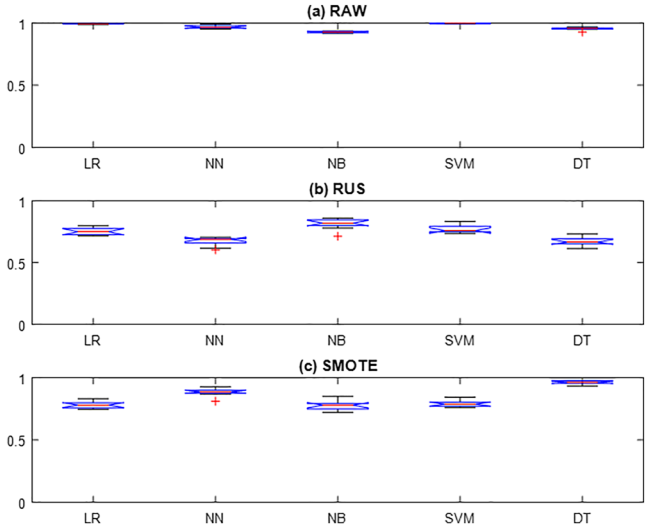

FIGURE 7 Box plots, sensitivity comparison

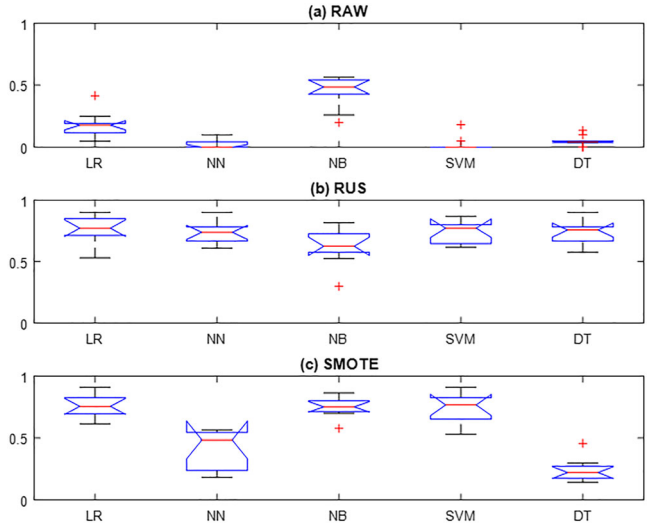

FIGURE 8 Box plot, specificity comparison

$$S_i = \frac{\sum_{k=1}^{K} s_{i\,k}\, AUC_{i\,k}}{\sum_{k}^{K} AUC_{i\,k}}, \quad i = 1, \cdots, I,\ k = 1, \cdots, K. \tag{1}$$

Figure 9 shows IF scores for each of 24 predictors which are listed in Table 4. The highest IF score noted is 0.984 for PAD, and the lowest IF score is for PCI, a value of 0.097. In the following, we used a fuzzy clustering approach for grouping these predictors based on IF scores.

Here, the predictors are clustered into three categories (high, medium, and low risk) factors based on IF score obtained earlier in this section. We have used fuzzy c-mean clustering (Bezdek, 1981; Chiu, 1994) to group the predictors into the three risk levels. This information is expected to be very useful for healthcare professionals for decision making at the admission into hospital for patients with ACS symptoms.

Amongst the predictors ranked in Section 5 using fuzzy c-mean clustering, seven predictors (PAD, CHF, Val_D, TIA, CVA_TIA, CABG, and Killip Class) have the highest impact towards in-hospital mortality due to ACS. Our findings are supported by a related research performed by Granger et al. (2003). These medical conditions are ranked as key risk factors for in-hospital mortality amongst ACS patients. Patti et al. (2007) have reported that PAD has a high impact to cause ACS. Dokainish et al.'s (2005) study provides empirical evidences for CHF as a major factor for ACS and mortalities due to ACS. Valvular disease (Val_D) is identified in Hasdai et al. (2003) as a salient cause of mortalities. Cardiovascular transient ischemic attack (CVA_TIA) impacts cardiac mortality and has been reported in Schwamm et al. (2009) along with including ACS as a cause of

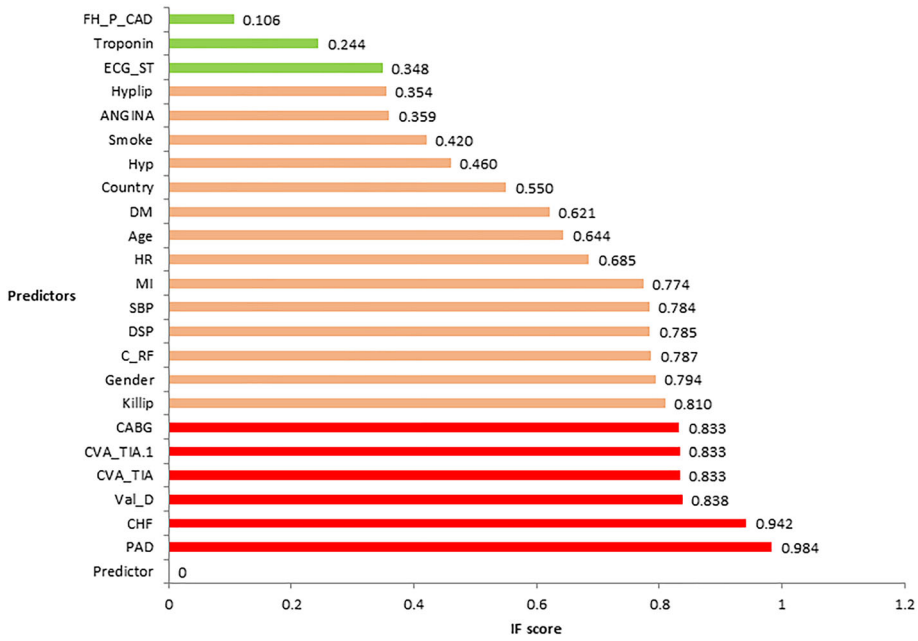

FIGURE 9 Information fusion score for predictors

death. Rastan et al. (2006) showed that there is a strong effect of CABG condition in patients' propensity suffering with ACS. Granger et al. (2003) in an evidence-based study showed that Killip class presentation is also an essential predictor for in-hospital mortality due to ACS.

The predictors related to demographic characteristics were found to be medium level predictors of in-hospital mortality. Gender and age were found to have an IF score of 0.794 and 0.644, respectively. Thus, gender is as strong a predictor as the medical conditions. A recent study by Dag et al., 2017 has underlined the importance of psychosocial predictors. In many cases, the demographic predictors may exhibit interaction effects with medical conditions in predicting mortality.

6 | CONCLUSIONS AND FUTURE RESEARCH SUGGESTIONS

The main objectives of this study include developing comprehensive, robust, data-driven approach to identify predictors (medical conditions) for in-hospital mortality using GRACE data for the Arabian Gulf. Unlike most previous studies in which the data mining and Machine Learning (ML) approaches are compared with to identify the best performing methods or predictive models based on commonly used performance measures such as accuracy, sensitivity, specificity. This study follows a recent trend of using a unification strategy of the predictive power of each of the data mining and ML approaches developed based on Information Fusion (IF) methodologies (Dag et al., 2017). The goal of unification predictive models is to uncover the importance, patterns, and relationships amongst predictors. Further, mitigation of prediction bias is very essential and has been achieved in this study using contemporary data imbalance management techniques, RUS, and SMOTE. Our approach contains six phases that systematically analyse the data using the state of the art tools from ML and advanced data analytics. Later, the unified IF scores for each of the predictors using the five (LR, SVM, NB, NN, and DT) models are classified into three (high, medium, and low) risk categories using a fuzzy c-mean clustering method. The study is also first of its kind because it considers the GRACE for Arabian Gulf Region (Granger et al., 2003) using a comprehensive data analytics framework from ML. In the analysis presented in this paper, we have answered the following:

1. How are ML methods combined with the traditional statistical methods to improve the overall prediction performance?
2. How can a decision support model be developed to help medical practitioners to predict individualized prediction outcomes for a patient or a group of patients with similar medical history (conditions)?
3. Which predictors (medical history records) are most essential to be considered for a patient by the medical experts to avoid adverse outcomes, that is, in-hospital mortality?
4. How can predictors be efficiently grouped with respect to their levels of risk of in-hospital mortality?
5. How can significant predictors be collectively identified by using contemporary multiple predictive ML models along with traditional statistical methods using an IF-based framework?

In order to Question 1, several ML models (SVM, NB, artificial NN, and DT) are compared along with the traditional LR. Question 2 is answered by quantifying the predictors based on their overall impact in predicting the in-hospital mortality for a patient with the onset of a medical history. Questions 3 and 4 are addressed by embedding a fuzzy c-mean clustering algorithm to group predictors into three essential clusters (high, medium, and low) risk level predictors. Finally, Question 5 is answered by improving the overall prediction power using IF process for each of the predictors.

In summary, our main contributions involve developing simple decision support tools using visualization methods for healthcare practitioners and expert decision makers. It is recommended that the latter exercise superior care and treatment for patients with symptoms identified in this research as high-risk factors for in-hospital mortality using proposed frameworks.

Even though this research outlines a comprehensive data analytics framework for the problem, the study could have possibly missed additional predictors of high importance such as treatments/care provided to avoid adverse outcomes (in-hospital mortality) during the patient's stay at the hospital suffering with ACS. This analysis can guide health-care experts to select the appropriate treatment, which can yield best outcomes.

The directions for future research include developing predictive models for propensity analysis over a longitudinal period for the three periods. These periods can be as follows: in-hospital, 6-month post discharge, and and 1-year post discharge (Dag et al., 2017). An interesting aspect would be to study the relationships amongst predictors using a sophisticated framework from ML such as Bayesian belief networks (Topuz, Zengul, Dag, Almehmi, & Yildirim, 2017; Vila-Francés et al., 2013) or fuzzy-set qualitative comparative analysis (Duman, Tozanli, Kongar, & Gupta, 2017).

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CONFLICT OF INTEREST

No conflict of interest to be disclosed.

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APPENDIX A

Acronym	Predictor	Remarks
Age	Patient's age	See Table 2 (main paper)
ANGINA	Angina pectoris	Medical history, Yes = 1, No = 0
C_RF	Chronic rental failure	Medical history, Yes = 1, No = 0
CABG	Coronary artery bypass grafting	Medical history, Yes = 1, No = 0
CHF	Congestive heart failure	Medical history, Yes = 1, No = 0
Country	Country of admission record	See Table 2
CVA_TIA	Cardiovascular transient ischemic attack	Medical history, Yes = 1, No = 0
CVA_TIA1	Transient ischemic attack (stroke)	Medical history, Yes = 1, No = 0
DM	Diabetes mellitus	Medical history, Yes = 1, No = 0
DBP	Diastolic blood pressure	See Table 2
ECG_ST	Electrocardiography elevation	

(Continued)

Acronym	Predictor	Remarks
		1 = ECG elevation, 2 = LBBB, 3 = T-waver inversion, 4 = Other, 5 = Normal
FH_P_CAD	Family history of premature coronary artery disease	Medical history, Yes = 1, No = 0
HR	Heart rate	See Table 2
Hyp	Hypertension	Medical history, Yes = 1, No = 0
Hyplip	Hyperlipidemia	Medical history, Yes = 1, No = 0
Killip	Killip class presentation	See Table 2
MI	Myocardial infarction	Medical history, Yes = 1, No = 0
PAD	Peripheral artery disease	Medical history, Yes = 1, No = 0
PCI	Percutaneous coronary intervention	Medical history, Yes = 1, No = 0
SBP	Systolic blood pressure	See Table 2
Smoke	Smoke history	Medical history, Yes = 1, No = 0
Troponin	Troponin level elevated	Medical history, Yes = 1, No = 0
Val_D	Valvular disease	Medical history, Yes = 1, No = 0